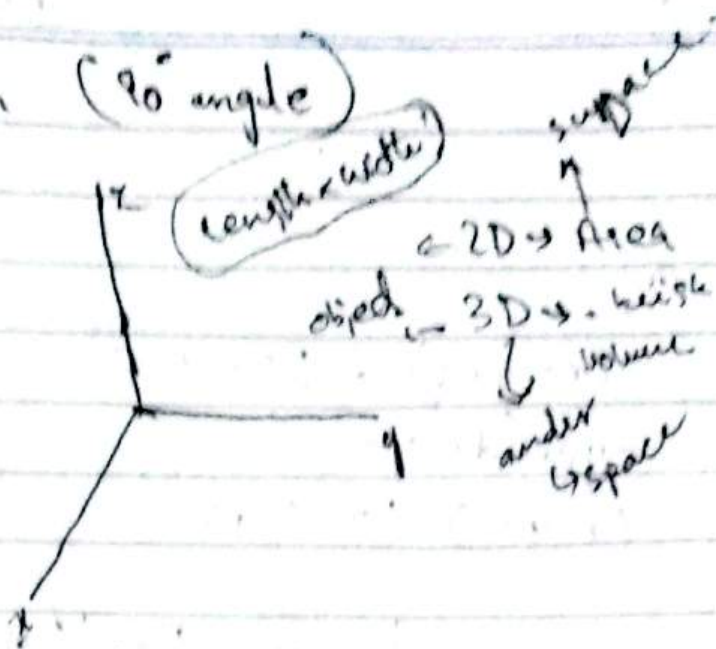


object

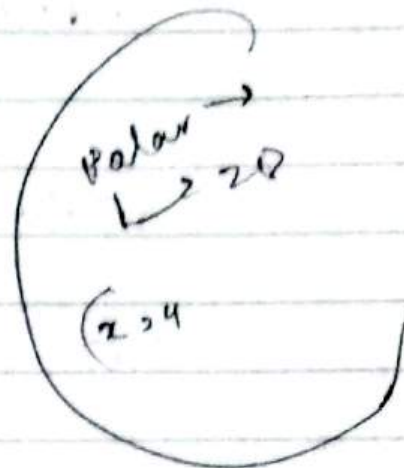
shadow on axis

3D coordinate System (90° angle)



2D → only one side at a time → e

3D → 2 side at a time



Ex.

cylinder

- tube
- 2 circular + 1 curved
- $\pi r^2 h$
- $2\pi r(h + r)$
- r, θ, z

spherical

- Ball
- Binned
- $\frac{4}{3}\pi r^3$
- $4\pi r^2$

Co-ordinate in 3D → rectangular/cartesian

$$(x, y, z) = (3, 4, 5)$$

$$\text{Polar} = r = \sqrt{x^2 + y^2}$$

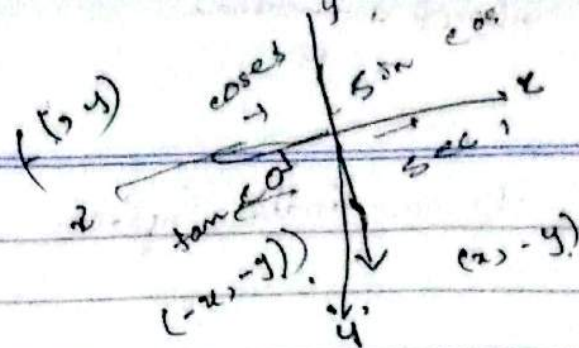
$$\theta = \tan^{-1}(y/x)$$

$$(x, y) = r(\cos \theta + i \sin \theta)$$

$$\text{Polar} = (r, \theta)$$

idr value
+ hi
karain
Dattani
rue

sum to 180
compl
90° sum



Q:

$$(x, y, z) = (3, 4, 5)$$

cylindrical

$$(r, \theta, z) = (r, \theta, z)$$

$$= \sqrt{x^2 + y^2}, \tan^{-1}(y/x), z$$

⊙ radian mai
w to pi man
convert karne
w pi

ca to cy

$$\sqrt{9+16}, \tan^{-1}(4/3)$$

$$= \sqrt{25}, \tan^{-1}$$

$$= 5, \theta = 53.13^\circ, 5$$

$$= (r, \theta, z) = (5, 53.13, 5)$$

$$x = r \cos \theta$$

$$x = 5 \cos 53.13^\circ$$

$$x = 5 \times 3/5 = 3$$

$$y = 5 \sin 53.13^\circ$$

$$y = 5 \times 4/5 = 4$$

$$z = 5$$

ca to cy

conditions.

$$\theta = \tan^{-1}(y/x)$$

$$\rightarrow \text{if } x > 0, y > 0 \\ \Rightarrow \theta \text{ (same)}$$

$$\rightarrow \text{if } x < 0 \text{ and } y > 0$$

$$\theta = \pi$$

$$\rightarrow \text{if } x < 0 \text{ and } y < 0 \\ \pi - \theta$$

$$\rightarrow \text{if } x > 0 \text{ and } y < 0$$

$$-\theta$$

Cartesian to cy:

$$\vec{r} = (x, y, z) = (-3, 3, 2)$$

$$= \sqrt{9+9}, \quad \theta = \tan^{-1}(3/3) \\ = \sqrt{18} \quad \begin{matrix} 45^\circ \rightarrow \theta = \pi \\ \rightarrow 180 - 45 = 135 \end{matrix}$$

$$= (3\sqrt{2}, 135^\circ, 2)$$

$$= (3\sqrt{2}, -135^\circ, 2)$$

$$x = 3\sqrt{2} \cos(135^\circ) =$$

$$3\sqrt{2} \cos(-135^\circ) = -3$$

$$y = 3\sqrt{2} \sin(135^\circ) = 3$$

$$z = 2 = 2$$

cy to ca

* Cylindrical to Cartesian

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Cartesian to Spherical

$$(x, y, z) = (\rho, \theta, \phi)$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}(y/x)$$

$$\phi = \cos^{-1}(z/\rho)$$

$$Q. (x, y, z) = (2\sqrt{3}, 6, -4)$$

$$\rho = \sqrt{(2\sqrt{3})^2 + (6)^2 + (-4)^2} = \sqrt{12 + 36 + 16} = \sqrt{64} = 8$$

$$\theta = \tan^{-1}\left(\frac{6}{2\sqrt{3}}\right) \text{ or } \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ = \frac{\pi}{3}$$

$$\phi = \cos^{-1}\left(\frac{-4}{2\sqrt{3}}\right) = \cos^{-1}\left(\frac{-2}{\sqrt{3}}\right)$$

$$\phi = \cos^{-1}\left(\frac{-4}{8}\right) = \cos^{-1}\left(\frac{-1}{2}\right) = 120^\circ \Rightarrow \frac{2\pi}{3}$$

$$(x, y, z) = (2\sqrt{3}, 6, -4)$$
$$(\rho, \theta, \phi) = \left(8, \frac{\pi}{3}, \frac{2\pi}{3}\right)$$

Spherical to cartesian

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\textcircled{a}: x = \rho \sin \phi \cos \theta$$

$$\begin{aligned} x &= 8 \sin(120) \cos(60) \Rightarrow 8 \sin(2\pi/3) \cos(\pi/3) \\ &= 8(\sqrt{3}/2)(1/2) = \boxed{2\sqrt{3}} \end{aligned}$$

$$\begin{aligned} y &= \rho \sin \phi \sin \theta \\ &= 8 \sin(2\pi/3) \sin(\pi/3) = 8(\sqrt{3}/2)(\sqrt{3}/2) \\ &= \boxed{6} \end{aligned}$$

$$\begin{aligned} z &= \rho \cos \phi \\ &= 8 \cos(30) = \end{aligned}$$

$$= 8(-1/2) = \boxed{-4}$$

Ques,

Q1. $(8, \pi/4, \pi/6)$ from spherical to cartesian

$$x = \rho \sin \phi \cos \theta =$$

$$x = 8 \sin(\pi/6) \cos \pi/4 \Rightarrow 2\sqrt{2}$$

$$y = \rho \sin \phi \sin \theta$$

$$y = 8 \sin(\pi/6) \sin(\pi/4) \Rightarrow 2\sqrt{2}$$

$$z = \rho \cos \phi$$

$$z = 8 \cos(\pi/4) = 4\sqrt{2}$$

Q2: $(1, \pi/2, 1)$ from cylindrical to spherical.

$$(r, \theta, z) = (1, \pi/2, 1)$$

$$(r, \theta, z) = (\rho, \theta, \phi)$$

For ρ

$$\rho = \sqrt{r^2 + z^2}$$

$$\rho = \sqrt{1^2 + 1^2}$$

$$\rho = \sqrt{2}$$

$$45 \rightarrow \pi/4$$

$$\theta = \tan^{-1}(y/z)$$

$$\theta = \tan^{-1}(\pi/2)$$

$$\phi = \cos^{-1}(z/\rho)$$

$$\phi = \cos^{-1}(1/\sqrt{2}) = 45^\circ = \pi/4$$

$$(r, \theta, z) = \rho(, \theta, \phi)$$

$$= (\sqrt{2}, \pi/2, \pi/4)$$

solved!

Q3: $(2, \pi/2, 1)$ from cylindrical to spherical

$$(r, \theta, z) = (\cancel{2}, \cancel{\pi/2}, \cancel{1}) \quad (\rho, \theta, \phi)$$

$$\rho = \sqrt{r^2 + z^2} = \sqrt{2^2 + 1} = \sqrt{5} \Rightarrow \rho = \sqrt{5}$$

$$\theta = \pi/2$$

$$\phi = \cos^{-1}(z/\rho) = \cos^{-1}(1/\sqrt{5}) = 63.43$$

$$63.43^\circ \times \pi/180 = 0.3524\pi$$

$$(\rho, \theta, \phi) = (\sqrt{5}, \pi/2, 0.3524\pi)$$

$$\phi = \cos^{-1}$$

$$(r, \theta, z)$$

$$\text{Q3. } 2nd$$

$$\theta$$
$$\phi$$
$$\rho$$

$$(1, \theta, z)$$

Q4: $(5, \pi/3, 2\pi/3)$ from spherical to cartesian and cylindrical

→ spherical to cartesian

$$(\cancel{5}, \pi/3)$$

$$(\overset{x}{5}, \overset{y}{\pi/3}, \overset{z}{2\pi/3})$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\rho = \sqrt{(5)^2 + (\pi/3)^2 + (2\pi/3)^2}$$

$$\rho =$$

$$x = \rho \sin \phi \cos \theta$$

$$x = 5 \sin(2\pi/3) \cos(\pi/3)$$

$$x = 5(\sqrt{3}/2)(1/2)$$

$$x = 5\sqrt{3}/4$$

$$z = \rho \cos \phi$$

$$y = 5 \sin \phi \sin \theta$$

$$y = 5(\sqrt{3}/2)(\sqrt{3}/2)$$

$$z = 5 \cos(2\pi/3)$$

$$y = 15/4$$

$$z = 5(-1/2)$$

$$z = -5/2$$

2) Spherical to cylindrical

$$r = \rho \sin \theta$$

$$r = 5 \sin(2\pi/3)$$

$$r = 5(\sqrt{3}/2)$$

$$r = \frac{5\sqrt{3}}{2}$$

$$\theta = \pi/3$$

$$z = \rho \cos \theta$$

$$z = 5 \cos(2\pi/3)$$

$$z = -5/2$$

Ans,